

GOKUL D

Embedded Systems Engineer | AI & IoT Developer

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PROFESSIONAL SUMMARY

Third-year Bachelor of Engineering (B.E.) student in Electronics and Communication Engineering with hands-on experience in embedded systems and firmware development using Embedded C across ESP32, STM32, and Arduino platforms. Skilled in building IoT and cloud-connected systems, with additional exposure to artificial intelligence, computer vision, and robotics through academic projects. Practical exposure to industrial and marine automation gained through internships, along with a published research paper on automation, machine learning, and embedded systems. Seeking an Embedded Systems, Firmware, IoT, or AI and Robotics internship to apply these skills to real-world engineering solutions.

TECHNICAL SKILLS

Programming Languages: Embedded C, C++, Python, Dart, Java, JavaScript

Embedded Systems & Microcontrollers: ESP32, STM32, Arduino

IoT & Cloud: Firebase, REST API, Flask, JSON

Artificial Intelligence & Computer Vision: OpenCV, OpenRouter API, LLM Integration, Computer Vision

Mobile Development: Flutter, Android Studio, Shared Preferences

Electronics & Hardware: Sensor Interfacing, PCB Design, Circuit Debugging, Control Panel Wiring, Marine Electronics

Development Tools: VS Code, Git, GitHub, Arduino IDE

Operating Systems: Windows, Linux, Ubuntu

EDUCATION

Bachelor of Engineering - Electronics and Communication Engineering | Adithya Institute of Technology, Coimbatore (Anna University)
2024 - 2028 | Third Year | CGPA: 8.38

INTERNSHIP EXPERIENCE

Assistant to Senior Service Engineer - INS Vikrant | Navicom Technology International Pvt. Ltd.

May 25, 2026 - June 05, 2026 | Marine Electronics, Navigation & Communication Systems

- Assisted the Senior Service Engineer with maintenance and routine servicing of navigation and communication equipment.
- Gained practical exposure to marine electronics systems associated with the Indian Naval Ship INS Vikrant at Karwar Naval Base.
- Supported technical activities related to communication equipment maintenance in a marine engineering environment.

Industrial Trainee - Marine Automation | SMEC Automation Pvt. Ltd.

June 05, 2025 - June 20, 2025 | Control Panel Building, Marine Automation

- Gained hands-on exposure to control panel building and electrical wiring within industrial automation systems.
- Interpreted electrical schematics to support panel assembly and wiring accuracy.
- Assisted with electrical testing procedures for control panels.
- Acquired practical understanding of marine control systems and industrial automation practices.

SELECTED ENGINEERING PROJECTS

AIR - Intelligent Classroom Assistant Using IoT and Cloud Technology

Project Leader / Embedded Systems Developer | ESP32-CAM, Arduino, Embedded C, Firebase, DFPlayer Mini, Ultrasonic Sensor, LCD

- Led development of an IoT-based classroom automation system, integrating ESP32-CAM for barcode-based attendance tracking.
- Interfaced ultrasonic sensors and an LCD display for real-time classroom monitoring and feedback.
- Implemented audio interaction using a DFPlayer Mini module for user notifications.
- Connected the system to Firebase for cloud-based data storage and built a web dashboard for attendance monitoring.

AMPET - AI-Based Smart Energy Theft Detection System

Project Leader / Embedded Systems Developer | ESP32, ACS712, SCT-013, ZMPT101B, Python, Flask, HTML, CSS, JavaScript

- Designed an ESP32-based system for real-time monitoring of current and voltage parameters using ACS712, SCT-013, and ZMPT101B sensors.
- Developed a Python and Flask backend to process and analyze acquired electrical data.
- Built a web-based interface using HTML, CSS, and JavaScript to visualize electrical parameters.
- Implemented a smart monitoring concept aimed at identifying potential energy theft patterns.

SenSafe - Smart Wearable Women Safety System

Project Leader | ESP32, BLE, Flutter, Embedded C

- Developed a wearable safety device using ESP32 with embedded firmware written in Embedded C.
- Implemented BLE communication between the wearable hardware and a companion mobile application.
- Built the companion mobile application using Flutter for user interaction and device connectivity.

SERA - Smart Electronics Research Assistant

Project Leader | Flutter, OpenRouter API, NVIDIA AI Models, REST API

- Built a Flutter-based mobile application to assist users with electronics-related research and learning.
- Integrated OpenRouter API and NVIDIA AI models to generate AI-assisted responses.
- Implemented REST API communication for reliable data exchange between the application and AI services.

VENOM - AI-Powered Intelligent Robotic Assistant

Hardware Engineer | Arduino, Servo Motors, PCA9685, OpenCV, AI

- Developed robotic hardware using Arduino for embedded control of the system.
- Implemented servo motor control via a PCA9685 driver for precise robotic movement.
- Integrated OpenCV and AI components to add visual perception capability to the robot.

Smart Parcel Management System

Project Leader | Python, OpenCV, QR Scanner, PyQt

- Developed a Python-based parcel management application using OpenCV for QR code scanning.
- Built a PyQt graphical interface for managing and tracking parcel records.

RESEARCH PUBLICATION

"Exploring the Synergy of Automation, Machine Learning, and Embedded Systems: Components, Applications, and Future Trends"

International Journal of Progressive Research in Engineering Management and Science (IJPREMS) | January 2025

CERTIFICATIONS & TECHNICAL ACTIVITIES

- German A1 Level Certification
- Industrial Training - Navicom Technology International Pvt. Ltd.
- Industrial Training - SMEC Automation Pvt. Ltd.
- National Level IPR Innovation Challenge 2026
- DataXscape National Level Hackathon
- Turtle BOT3 ROS Workshop
- Research Publication Certificate - IJPREMS

ACHIEVEMENTS

- Published research paper, "Exploring the Synergy of Automation, Machine Learning, and Embedded Systems," in IJPREMS (January 2025).
- Participant, National Level IPR Innovation Challenge 2026.
- Participant, DataXscape National Level Hackathon.
- Recipient, Young High Flier Award.
- German A1 Language Certification.
- Participant, Turtle BOT3 ROS Workshop.
- Completed Marine Automation Industrial Training at SMEC Automation Pvt. Ltd.
- Completed Marine Electronics Industrial Training at Navicom Technology International Pvt. Ltd., with exposure to navigation and communication equipment associated with INS Vikrant.

AREAS OF INTEREST

Embedded Systems, Internet of Things (IoT), Artificial Intelligence, Robotics, Edge AI, Marine Automation, Firmware Development, Computer Vision, Smart Energy Systems, Industrial Automation